Bridging the Intertemporal Domain Shift for Remote Sensing Change Detection via Domain Adaptation

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***Abstract*—Detecting changes by comparing two remote sensing images of the same area, acquired at different times, is often complicated because these images are typically captured under varying acquisition conditions, including but not limited to different seasons and inherent differences in imaging spectral characteristics. This discrepancy creates a "temporal domain shift," leading to errors such as false alarms and blurred boundaries in results. To address this challenge, this article proposes an adversarial domain adaptation network for change detection (ADANet-CD). ADANet-CD employs a two-stage progressive optimization strategy. In the encoding stage, it first constructs a dynamic adversarial mechanism between the feature extractor and the domain classifier. At the same time, a temporal mutual-aware attention module (TMAM) is designed to enhance domain-invariant feature discrimination by leveraging the fusion and interaction of spatiotemporal features. Then, a binary-guided distance loss (BDL) based on Minkowski distance is introduced to explicitly constrain the distribution alignment of bi-temporal features. In the decoding stage, an interactive difference enhancement module (IDEM) is proposed. This module aims to sharpen the model's focus on true changes while reducing its reliance on domain-specific details, further mitigating the domain shift. Experimental results on the CDD, LEVIR-CD, and DSIFN-CD datasets demonstrate that ADANet-CD outperforms existing methods in terms of both precision and *F*1-score, exhibiting superior performance in both change region localization and nonchange region suppression.**

***Index Terms*—Bi-temporal alignment, change detection (CD), domain adaptation, temporal mutual-aware.**

I. Introduction

REMOTE sensing change detection (CD) is a vital application of remote sensing technology. It precisely identifies and quantifies changes in land cover and status. This is done by analyzing remote sensing images of the same area taken at different times. This technology is crucial for monitoring urban development [1], [2], [3], [4], assessing natural disaster damage [5], [6], [7], managing agricultural resources [8], [9], militarizing reconnaissance [10], [11], and achieving global sustainable development goals. It provides timely and reliable data for decision-making.

Researchers have proposed various remote sensing image CD methods to improve accuracy. These are mainly traditional and deep learning-based methods. Traditional methods extract features from image pixels, textures, and shapes. They are divided into pixel-level and object-level approaches. Pixel-level methods compare images pixel by pixel across different times. Examples include image algebra [12], [13], [14], image transformation [15], [16], [17], and classification [18]. Object-level approaches use image segmentation to group pixels into object units. This helps handle complex texture details. Common approaches include direct object CD and postclassification object CD [19], [20]. However, traditional methods are sensitive to noise and spectral variations, showing poor robustness. They also struggle to capture deep semantic features and spatial information, limiting their effectiveness in complex situations. Furthermore, traditional methods rely on manually set thresholds, have limited feature expression, and lack adaptability and generalization.

To overcome traditional method limitations, deep learning-based remote sensing image CD methods have emerged as a research frontier. Deep learning methods, especially convolutional neural networks (CNNs), show strong robustness and generalization. They automatically extract high-level semantic features from images. This is due to their power in modeling nonlinear relationships and extracting multilevel features. This offers a new technical path for CD in complex scenarios. For instance, Zhang et al. [21] combined CNNs and graph neural networks. This allowed for more flexible modeling of complex object shapes. They updated node information through internode message passing for CD. Ye et al. [22] improved traditional CNNs by adding 3D convolutions. This extracted spatial features and temporal change information from remote sensing images, reducing the semantic gap in deep CNN features. Dong et al. [23] proposed ConMamba, a hybrid architecture combining CNNs and state-space models (SSMs). They integrated CNNs and SSMs through a feature interaction module. This built a high-performance hybrid encoder that extracts local details and integrates global dependencies. This overcomes the high computational complexity of traditional Transformers and the single Mamba architecture limitations in local information extraction. These methods have made significant progress in feature extraction. They

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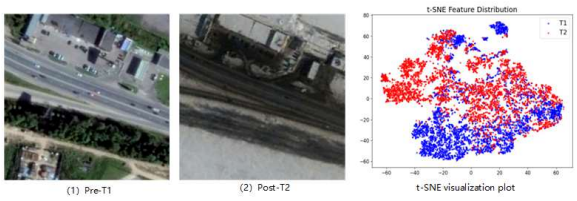


Fig. 1. t-SNE feature distribution visualization of bi-temporal remote sensing image pairs within the dataset.

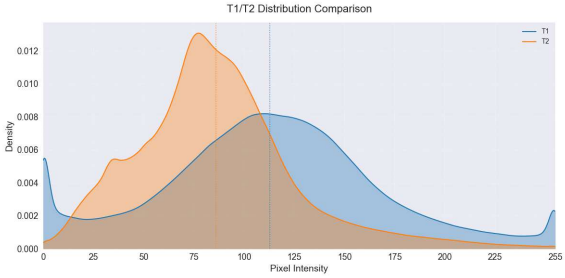


Fig. 2. Feature distribution of bi-temporal image pairs in the dataset.

provide powerful feature expression for CD tasks. Zhang et al. [24] also proposed a feature-prediction alignment-based semi-supervised CD method. This method uses unlabeled remote sensing images through class-aware feature alignment and pixel prediction alignment strategies. It addresses the excessive reliance of traditional fully supervised methods on pixel-level annotations. It also deals with the uncontrollable optimization direction and the lack of physical meaning in existing semi-supervised methods.

Existing deep learning-based CD methods usually focus on extracting deep semantic features and enhancing feature representation. However, they implicitly assume spatial consistency in bi-temporal image feature distributions. They do not fully consider the temporal domain shift between bi-temporal remote sensing images. Figs. 1 and 2 show feature visualizations using t-SNE and JS divergence distributions. The analysis indicates a clear domain shift in the feature space of bi-temporal image data.

In recent years, domain adaptation techniques have been widely used in computer vision. They effectively address the domain shift problem in deep learning. These techniques have been successfully applied to tasks like object detection [25], [26], semantic segmentation [27], and image classification [28]. Their goal is to solve inconsistencies in feature distribution within datasets. The main goal of domain adaptation is to reduce or remove feature distribution differences between domains. This bridges the gap between domains. Ultimately, it allows deep neural networks to learn more discriminative and domain-invariant feature representations.

Therefore, effectively solving the critical problem of intertemporal data domain shift is a core challenge. This challenge needs breakthroughs in remote sensing image CD. This builds on the strong foundation of deep semantic feature extraction.

Based on this analysis, this article proposes an adversarial domain adaptation-based remote sensing image CD method. This method aims to explicitly reduce feature distribution differences between bi-temporal images. It achieves alignment in the feature space. This effectively lessens the domain shift problem and improves the accuracy and stability of CD. The main contributions of this article are as follows.

1) We proposed a novel adversarial domain adaptation network for CD (ADANet-CD) for bi-temporal remote sensing images. This model uses an adversarial training mechanism between the feature extractor and the domain classifier through gradient reversal. This aligns bi-temporal image features into a shared space. This effectively reduces interference in CD from external environmental factors like differences in imaging spectral characteristics and seasonal variations.
2) To further help domain adaptation with feature alignment between bi-temporal images, this article designs a binary-guided distance loss (BDL), based on Minkowski distance, and a temporal mutual-aware attention module (TMAM) between the extracted bi-temporal features. This design strengthens bi-temporal feature representation. More importantly, it optimizes feature distribution to further enhance the domain adaptation's alignment effect, promoting effective feature alignment.
3) This article innovatively proposes an interactive difference enhancement module (IDEM). This module aims to increase the model's focus on change information and reduce its reliance on domain-specific information. This improves the precise localization of change regions. It also significantly enhances the detection accuracy of edge features, boosting the model's overall performance.

The rest of this article is organized as follows. Section II reviews related research on remote sensing image CD and domain adaptation. Section III explains the proposed overall model framework and analyzes each module. Section IV validates the method's effectiveness through experiments and discusses the results. Section V concludes this article and offers future research directions.

II. Related Work

A. Deep Learning-Based CD

Currently, deep learning-based CD methods are dominant and a research hotspot in remote sensing. These methods use neural networks with powerful feature extraction to get deep features from remote sensing images. Daudt et al. [29] introduced the first CNN-based CD models: FC-EF, FC-Siam-Conc, and FC-Siam-Diff. These models use a fully convolutional fusion network. They stack bi-temporal remote sensing images and use the U-Net's skip connections to fuse multiscale features. This initially proved deep learning's effectiveness in CD. However, these early methods rely heavily on input data quality and are sensitive to noise and change types. They also use contextual and temporal information. This leads to limited detection accuracy in complex situations. To improve feature extraction, Fang et al. [30] proposed a densely connected Siamese network for CD. This method deeply fuses

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and extracts multilevel features through an encoder–decoder structure. This effectively improves the ability to capture change features. Chen and Shi's STANet model [31] adds a spatiotemporal self-attention mechanism to enhance feature representation. This allows it to dynamically focus on features from different time phases and spatial locations, improving CD flexibility and accuracy. Fang et al. [32], in the Changer model, focus on enhanced feature extraction and fusion strategies. This aims to better capture subtle changes and change features in complex scenarios. While these deep learning methods show advantages in capturing local structural information, they lack global feature modeling capabilities and have poor generalization performance.

To further improve model generalization and overcome CNNs' weaknesses in long-distance dependency modeling, researchers are now adding the Transformer architecture to remote sensing image CD. Chen et al.'s BIT model [33] uses Vision Transformers for CD. It effectively models contextual information with a token mechanism. It also efficiently extracts and fuses multilevel features of bi-temporal remote sensing images. Bandara and Patel's ChangeFormer model [34] directly uses the Transformer architecture for CD. It effectively captures global features and changes relationships through the self-attention mechanism. These Transformer-based methods combine local features with global context modeling, offering a more complete feature representation. However, they still lack effective alignment of bi-temporal features during feature extraction. This makes it hard for the model to capture deep correlation information between bi-temporal images, affecting CD accuracy and robustness.

Based on this, some studies now explore using feature alignment to reduce feature differences between domains. This aims to lessen the domain shift phenomenon. Zhang et al.'s Bifa model [35] combines self-attention mechanisms and optical flow concepts. This aligns bi-temporal remote sensing images at different feature levels. It solves false detections caused by imaging condition differences in high-resolution remote sensing image CDs. Tian et al. [36] proposed learning feature map offsets and combining them with super-resolution reconstruction. This solves problems of inconsistent image resolution and alignment errors in multiresolution optical remote sensing image CDs. Zhou et al. [37] proposed a dual-granularity feature alignment (DgFA) method. This explicitly aligns bi-temporal remote sensing image features by integrating semantic alignment (for appearance heterogeneity of the same type objects) and spatial alignment (for affine geometric differences). This solves pseudo-changes caused by these differences in CD. Liu et al. [38] addressed unsupervised multimodal CD by proposing the superpixel-based autoencoder Kolmogorov–Arnold network (AEKAN). This method effectively learns and extracts modality-independent commonality features. Operating in an unsupervised framework, it aligns the feature spaces from different modalities. Subsequently, it quantifies the differences in these aligned features to achieve accurate CD. Building upon this work, they further introduced the CFRL [39] framework to tackle the feature alignment challenges in heterogeneous multimodal CD. This framework leverages a Siamese network encoder and two decoders. Modality alignment is achieved through a pseudo-multimodal image reconstruction process involving decoder swapping. In an unsupervised manner, CFRL extracts modality-independent common features from heterogeneous bi-temporal images and utilizes their differences for precise CD. These methods are designed to bridge significant modality gaps, such as between optical and SAR imagery, where sensing mechanisms and statistical distributions differ fundamentally. In contrast, the present work focuses on domain shift in homogeneous modalities—specifically, dual-temporal optical images subject to seasonal, illumination, and atmospheric variations. Although the distributional discrepancy in such scenarios is generally smaller, it remains a prevalent source of false alarms in real-world CD applications. Importantly, while multimodal methods aim for complete modality invariance, our approach recognizes that meaningful change signals may reside in subtle temporal variations. Therefore, we advocate for a more nuanced alignment strategy that mitigates nuisance variations without suppressing genuine changes—a balance that is particularly critical when changes are sparse and localized. However, in CD tasks, change regions are usually a small part of the image. If existing feature alignment methods mainly align invariant features, which are a larger proportion, this can lead to extensive alignment regions. In this case, overly aggressive alignment (over-alignment risk) might not effectively distinguish domain shift differences from real changes. This could incorrectly align real change regions, suppressing or eliminating real change signals, and ultimately leading to missed key change information.

Therefore, to effectively solve these problems, this article studies intertemporal domain adaptive CD methods. It maps bi-temporal image features from different domains into the same shared feature space. This effectively reduces the widespread domain shift problem between bi-temporal remote sensing images, achieving effective alignment of deep features between bi-temporal images.

B. Domain Adaptation for Remote Sensing Image CD

Domain adaptation (DA) is a key part of transfer learning. Its main goal is to effectively solve data distribution differences between a source domain and a target domain. Researchers have proposed many DA methods to address the domain shift problem. For example, Ganin and Lempitsky [40] introduced the maximum mean discrepancy (MMD) metric. This method minimizes the mean difference between source- and target-domain features to match their distributions. Sun and Saenko [41] proposed using the correlation alignment (CORAL) distance to align different domains. CORAL's core is to minimize differences in second-order statistics between domains. This achieves efficient domain adaptation without target-domain labels. Building on this, Pu et al. [42] used multikernel MMD (MK-MMD) as a constraint for domain adaptation, aligning features from different source data. Unlike methods based on statistical distance, Ganin et al. [43] proposed the domain-adversarial neural network (DANN). This creatively applied the adversarial training concept from generative adversarial networks (GANs). It uses a gradient reversal technique to dynamically adjust the feature extractor. A domain adversarial

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discriminator is designed and placed after the feature extractor. The discriminator aims to identify the domain of the input feature. The feature extractor's goal is the opposite: to create confusing, indistinguishable domain-invariant features. This makes it hard for the discriminator to identify the feature's source of the domain. This clever adversarial training allows the deep learning model to learn robust, cross-domain generalizable domain-invariant features. This significantly improves model performance in complex cross-domain tasks. Kundu et al. [44] proposed a new unsupervised domain adaptation method for a general source domain. This method transfers knowledge from a labeled source domain to an unlabeled target domain without source domain samples. It adapts to various category differences. Its core idea is to train a source domain model that can identify and reject samples outside the source distribution. This achieves domain adaptation when source domain data is absent. Hou and Zheng [45] performed style transfer by matching batch normalization statistics of generated source-style image features with features used for image translation stored in a pretrained source model. This achieved source-free domain adaptation.

Domain adaptation has become widely used in computer vision. However, in remote sensing image CD, research on DA is relatively limited. Applying DA to the remote sensing image CD can effectively reduce the domain shift problem. It also improves detection accuracy and reliability. Based on task objectives, DA in remote sensing image CD can be divided into three types: bi-temporal domain adaptation, multisource domain adaptation, and cross-regional domain adaptation. Bi-temporal domain adaptation solves domain shift caused by factors like illumination and seasonal changes. It aligns the feature distribution of bi-temporal images. For example, Ji et al.'s DA-IDANet [46] uses MMD as a domain adaptation constraint. It maps bi-temporal remote sensing images into a unified deep feature space for feature alignment. This reduces feature distribution differences and pseudo-change interference from factors like illumination changes and cloud occlusion. Multisource-domain adaptation addresses feature distribution differences between data from different sensors. It improves multisource data CD performance through feature alignment. For example, Zhang et al. [47] used MK-MMD as a domain adaptation constraint for feature space alignment. This solved the inconsistent feature distribution in heterogeneous optical and SAR remote sensing image CDs. It also improved model robustness. Cross-regional domain adaptation handles domain shift between remote sensing images from different geographical regions. This is where knowledge learned by the model on the source domain is hard to transfer to the target domain. Cross-regional domain adaptation improves model generalization by aligning the feature distribution of cross-regional images. Xiong et al. [48] proposed a wavelet Siamese network with a semi-supervised contrastive domain adaptation strategy to address the domain shift problem in remote sensing image CD. The method partitions image areas into three categories and applies a contrastive loss to effectively reduce domain discrepancy in unchanged regions while enhancing the discriminability of changed regions, thereby mitigating pseudo-changes. Tang et al.'s segmentation map difference-based domain adaptation method [49] uses MMD and Ada-UNet. It reduces feature distribution differences between datasets through feature alignment, channel attention, and segmentation map difference constraints during pretraining. This improves cross-domain CD accuracy and robustness.

However, the methods above all use statistical distance measurements for domain adaptation between bi-temporal images. This approach is usually global. It struggles to capture local or complex distribution differences. Its performance heavily depends on the chosen kernel function. While MK-MMD can somewhat alleviate this with multikernel methods, it still requires careful design and adjustment of hyperparameters. Additionally, statistical distance measurement methods are static during training. It is hard to dynamically adjust to changes in data distribution.

III. Proposed Method

A. Overall Network Architecture

The overall architecture of our proposed ADANet-CD model is shown in Fig. 3. The entire framework operates in an end-to-end manner. It tightly integrates data processing, feature learning, domain adaptation, difference enhancement, and final prediction. Our core design philosophy is to systematically address bi-temporal domain shift. The goal is to achieve high-precision CD. The process begins with the feature extractor, which utilizes a pretrained ResNet-50 network, to convert bi-temporal images into initial deep feature representations. This lays a solid foundation for all subsequent modules. These features then proceed to the core optimization stage: the implementation of the domain adaptation strategy. This strategy employs adversarial training between the feature extractor and a domain classifier. GRL forces the feature extractor to learn domain-invariant features. These features are insensitive to bi-temporal domain differences. This step is the cornerstone of the entire framework. It ensures that features from different periods are aligned in a shared deep feature space. This significantly mitigates pseudo-changes caused by bi-temporal domain shift. Building on this, the TMAM forms a strong synergy with the domain adaptation strategy. The TMAM is not an isolated component. It acts directly on the features processed by the domain adaptation. It uses a dual-branch attention mechanism. This facilitates deep interaction between T1 and T2 features, which enhances their discriminability. The TMAM's contribution provides higher-quality input to the GRL and domain classifier. This further optimizes the interdomain alignment. The output of the TMAM serves as the precise input for the BDL loss function. The BDL's objective is to impose a refined constraint on the distance between the TMAM-enhanced bi-temporal features. It minimizes the feature distance for unchanged regions to strengthen temporal consistency. Simultaneously, it maximizes the feature distance for changed regions to highlight differences. In this stage, BDL acts as a key auxiliary supervisor. It provides pixel-level, fine-grained constraints for the domain adaptation process. This ensures the model accurately quantifies and identifies change information while achieving domain alignment. Finally, the IDEM receives the features jointly optimized by the preceding modules. The IDEM's focus is to enhance the model's

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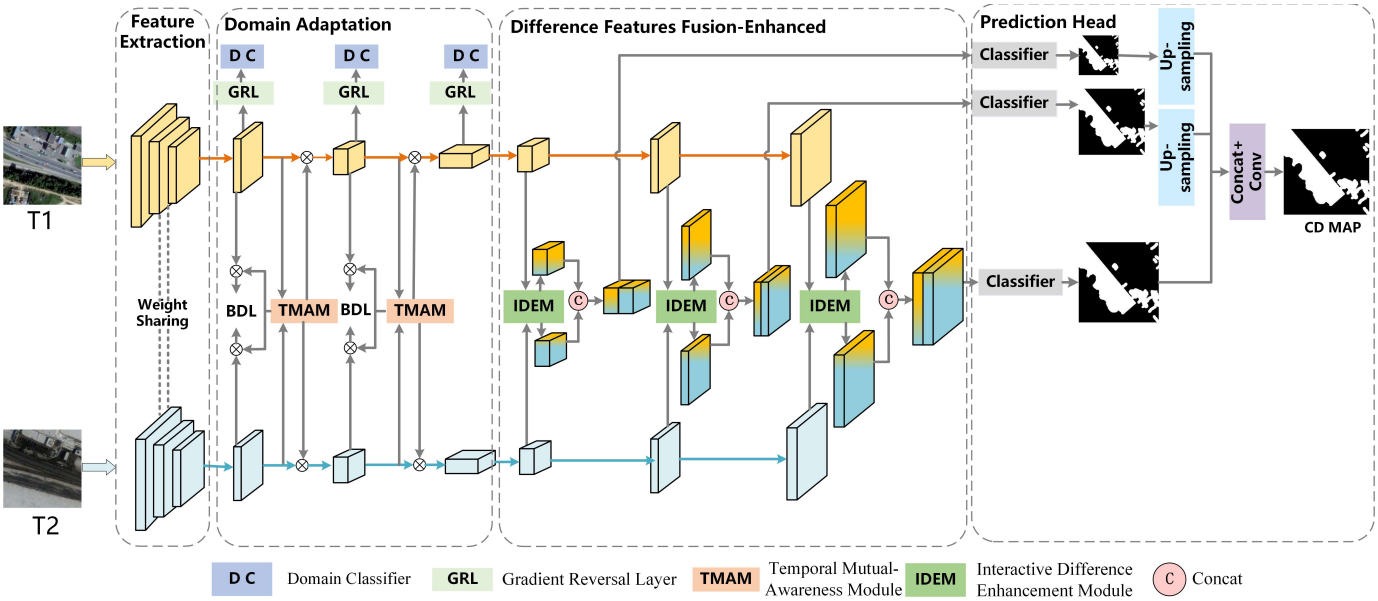


Fig. 3. Architecture of the proposed ADANet-CD. Bitemporal image features are extracted for the pretrained ResNet-50, and feature alignment is performed by domain adaptive constraints. The TMAM enhanced domain-invariant feature discrimination by leveraging the fusion and interaction of spatiotemporal features. Then, the IDEM is designed to sharpen the model's focus on true changes while reducing its reliance on domain-specific details, further mitigating the domain shift. Finally, the final binary variation map is obtained by going through the classifier.

attention to change information. It also reduces its reliance on domain-specific details. Through its unique interactive mechanism, the IDEM dynamically captures and amplifies subtle differences between the bi-temporal features. This guarantees that even in a highly aligned feature space, genuine change signals are clearly highlighted and accurately identified. The difference features processed by the IDEM are then fed into a classifier, which generates the final binary change map.

B. Domain Adaptation

In the encoding stage, this study builds on DANN. It creates an adversarial training mechanism between the feature extractor and the domain classifier using GRLs. This aims to map bi-temporal remote sensing image features into the same deep space. This effectively reduces the domain shift problem caused by external factors like differences in imaging spectral characteristics and seasonal variations. The main function of GRL is to guide the model to learn domain-invariant features. This achieves feature alignment between bi-temporal images, improving CD performance.

Specifically, as seen in Fig. 3, GRLs are placed after the feature extractor and linked to the domain classifiers. During forward propagation, GRL acts as an identity mapping, directly passing feature maps from the feature extractor to the domain classifiers. The classifiers then try to tell the source of the features (T1 or T2 branch). During backward propagation, GRL uses a gradient reversal mechanism. It multiplies the gradient from the domain classifiers by a negative coefficient, reversing the gradient direction. This creates an adversarial training relationship between the feature extractor and the domain classifiers. The feature extractor aims to generate features that the domain classifiers cannot distinguish as the source of. The domain classifiers aim to distinguish the feature source as accurately as possible. This adversarial mechanism forces the feature extractor to learn domain-invariant features that are not sensitive to temporal differences. These features effectively capture common information between bi-temporal images. They also preserve significant differences between changed and unchanged regions. As adversarial training continues, the domain classifiers gradually lose the ability to distinguish feature sources. This shows that the feature extractor has successfully aligned the bi-temporal image features into the same deep space. This process significantly reduces interference from temporal difference factors like differences in imaging spectral characteristics and seasonal variations on CD results. It enhances the model's robustness and generalization in complex scenes.

1) Temporal Mutual-Aware Attention Module: To further enhance the performance of domain adaptation techniques in CD models, this article proposes a TMAM to achieve more efficient feature alignment and domain adaptation for bi-temporal remote sensing imagery, as shown in Fig. 4. The TMAM is designed to precisely capture complementary and discriminative temporal-aware information from bi-temporal feature maps F_{T1} and F_{T2}. Its internal attention mechanism adopts a cascaded and fused design. First, for each input temporal feature map, the TMAM generates a corresponding channel attention map (M_C^{T1} and M_C^{T2}). This adaptively weights channels to highlight important semantic information. These channel-weighted features are then fused with the original features. They are subsequently fed into the spatial attention mechanism to generate corresponding spatial attention maps (M_S^{T1} and M_S^{T2}). The spatial maps capture spatial dependencies and focus on regions most significant for the CD task. Finally, by sequentially applying and fusing channel and spatial attention, the TMAM module generates a final pair of

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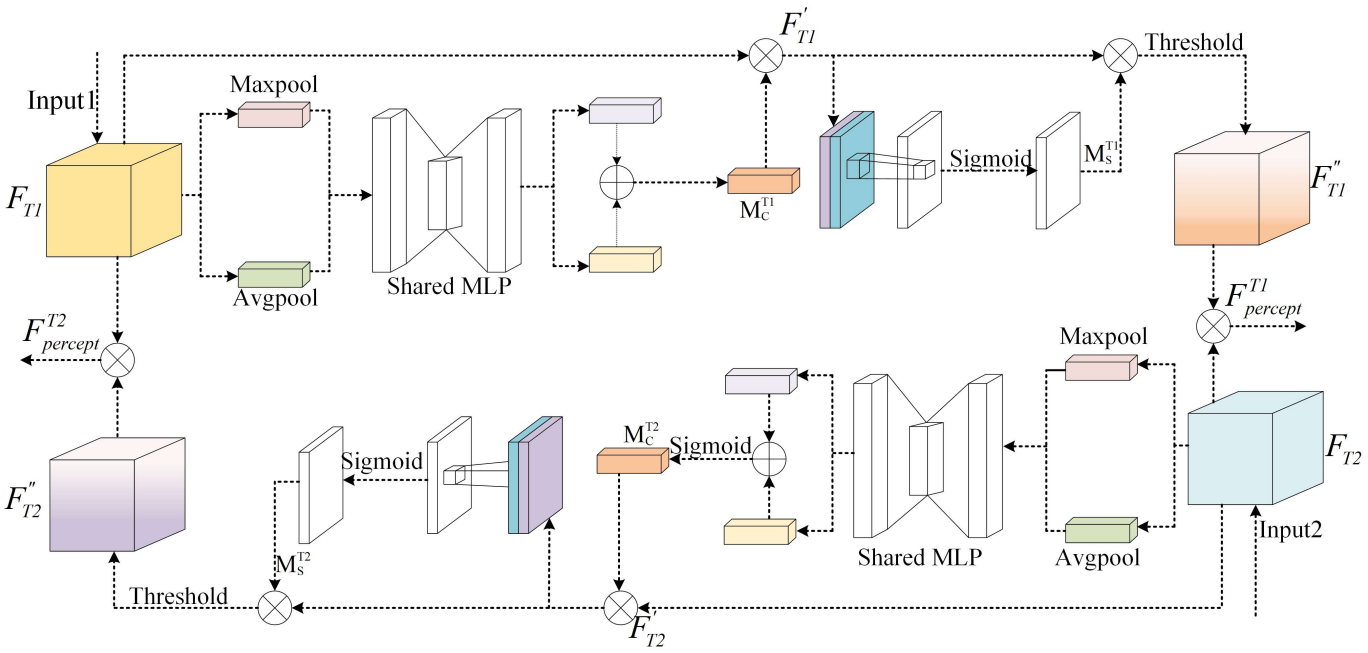


Fig. 4. TMAM.

perceptual feature maps (F^{T1}_{percept} and F^{T2}_{percept}). These maps are an enhanced representation highly fused with temporal context, channel semantics, and spatial saliency. They effectively suppress background noise and domain-specific information. This provides robust and discriminative domain-invariant features, establishing a solid foundation for subsequent CD tasks.

The TMAM module allows the T1 branch to perceive the T2 branch's feature information. This enhances temporal feature interaction. It also uses channel and spatial attention together, capturing salient information across both dimensions. The addition of a thresholding mechanism further lessens noise and irrelevant background effects. This improves the robustness and discriminative power of the feature representations. During domain adaptation, TMAM-enhanced features provide more discriminative representations for the gradient reversal layer (GRL) and domain classifiers. This multilevel feature processing greatly improves the model's ability to detect true changes. It also enhances its robustness against domain shift factors like seasonal variations. This offers a promising new direction for remote sensing image CD.

The mathematical expression of the TMAM can be defined as

$$M_c\left(F_{T1}\right)=\sigma\left(\text{MLP}\left(\text{AvgPool}\left(F_{T1}\right)\right)+\text{MLP}\left(\text{MaxPool}\left(F_{T1}\right)\right)\right) \tag{1}$$

$$F^{'}_{T1}=M_c\left(F_{T1}\right)\otimes F_{T1}. \tag{2}$$

M_c represents the channel attention weights, denotes the given input feature map of the T1 branch, AvgPool() and MaxPool(), respectively, represent the global average pooling and global max-pooling operations, through which the spatial dimensions of the input feature are compressed to 1, generating two channel descriptors. MLP is a shared multilayer perceptron. The channel descriptors undergo feature transformation through the MLP to generate the channel attention map. The values of the channel attention map are normalized to the range [0, 1] using the Sigmoid function $\sigma()$. Finally, the channel attention map is element-wise multiplied with the original feature map to obtain the channel attention weighted feature map $F^{'}_{T1}$

$$M_s\left(F^{'}_{T1}\right)=\sigma\left(f^{7\times7}\left(\left[\text{AvgPool}_c\left(F^{'}_{T1}\right),\text{MaxPool}_c\left(F^{'}_{T1}\right)\right]\right)\right) \tag{3}$$

$$F''_{T1}=T\left(M_s\left(F^{'}_{T1}\right)\otimes\left(M_c\left(F_{T1}\right)\otimes F_{T1}\right)\right). \tag{4}$$

M_s represents the spatial attention weights. AvgPool() and MaxPool(), respectively, represent the global average pooling and global max-pooling operations, generating two spatial descriptors. The two spatial descriptors are concatenated along the channel dimension, and then a spatial attention map is generated through a convolution layer. After passing through the sigmoid activation function, the obtained spatial attention weights are applied to the intermediate feature map $F^{'}_{T1}$. Subsequently, the feature map processed by the dual attention mechanism undergoes the calculation of an adaptive threshold function to obtain the final enhanced feature map F''_{T1}, with the operation for the T2 branch being the same as for the T1 branch.

The enhanced feature map of each branch is element-wise multiplied with the original feature of the other branch to perform intertemporal feature interaction, enabling each branch to autonomously perceive the features of the other, thereby promoting feature alignment between bi-temporal images and improving the domain adaptation capability of the model. The mathematical formula can be expressed as

$$F^{T1}_{\text{percept}}=F_{T2}+F''_{T1} \tag{5}$$

$$F^{T2}_{\text{percept}}=F_{T1}+F''_{T2}. \tag{6}$$

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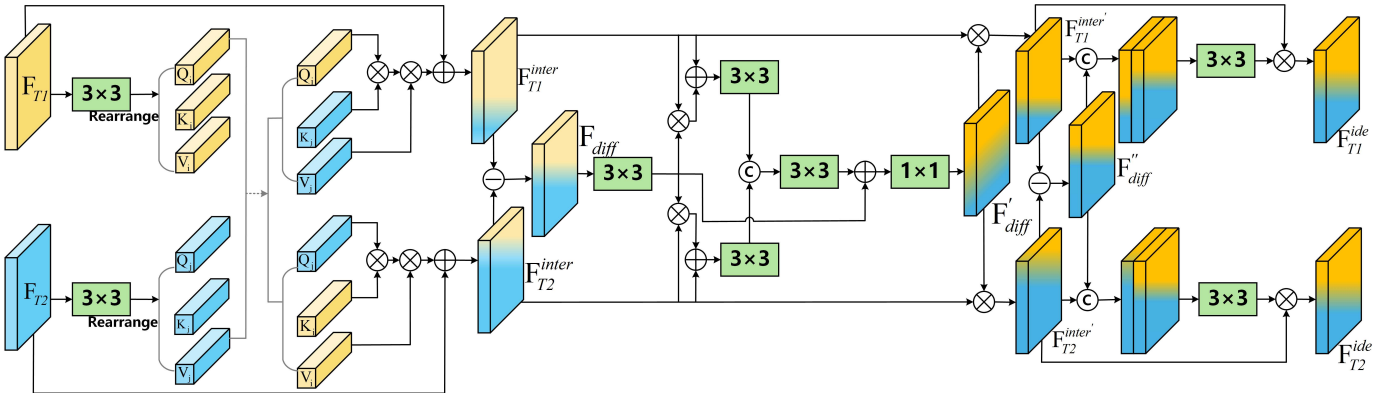


Fig. 5. IDEM.

F^{T1}_{percept} represents the perceptual feature interaction map of the T1 branch and T2 branch. F^{T2}_{percept} represents the perceptual feature interaction map of the T2 branch and T1 branch.

2) Binary-Guided Distance Loss: This study proposes a BDL. It builds on the TMAM feature enhancement. BDL guides the model to learn consistent information in bi-temporal remote sensing images. It also helps distinguish changed areas. The core idea of BDL is that features from T1 and T2 phase images at the same location should contain both shared, domain-agnostic information and unique information. For unchanged areas, T1 and T2 features, after removing specific information, should be very consistent. This means a small feature distance. For changed areas, T1 and T2 features will show big differences. This means a large feature distance.

Specifically, BDL guides model training by calculating the distance between feature vectors at corresponding pixel locations in the TMAM-enhanced T1 and T2 phase feature maps. BDL's design aims to minimize the bi-temporal feature distance in unchanged areas. This prompts the model to focus on and learn shared information in phase-specific features. At the same time, for changed areas, BDL encourages the model to extract features that show temporal differences, resulting in a larger distance. Thus, BDL helps the model learn feature representations that capture commonality between bi-temporal images and distinguish temporal differences. This improves CD accuracy. BDL is defined as

$$L_{\text{BDL}} = \alpha \sum_{n=1}^{2} \text{dist}\left(F^n_{T1} \times A, F^n_{T2} \times A\right) \tag{7}$$

where dist() represents the Minkowski distance function, α is a tuning factor, F^n_{T1} is the feature map from the T1 branch, F^n_{T2} is the feature map from the T2 branch, n denotes the index of the feature map within the overall framework, A denotes the final perceptual feature map pair, and F^{T1}_{percept} percept and F^{T2}_{percept}, which are output by the TMAM module. This output is generated after cascaded processing and fusion by channel attention (M_C) and spatial attention (M_S). These feature maps are a deep fusion of bi-temporal information and refined attention mechanisms. They effectively highlight domain invariance and filter out irrelevant information. The BDL loss then minimizes the distance between these perceptual feature maps. This ensures that features of changed and unchanged regions are clearly separated in the target domain. This process ultimately enhances the model's domain adaptation capabilities.

C. Interactive Difference Enhancement Module

In bi-temporal remote sensing image CDs, change regions are usually small and scattered. This limits the accuracy of current methods. To better capture weak change information and precisely locate change regions, this article proposes an IDEM, shown in Fig. 5. The IDEM aimed to increase the model's focus on change information and reduce its reliance on domain-specific information. It does this by analyzing and modeling the dynamic changes and interactions of features at two time points. Specifically, the IDEM uses an attention mechanism to dynamically find and focus on potential change regions. It prioritizes enhancing feature representation in these regions. It also suppresses interference from irrelevant information, like static backgrounds. This targeted difference feature enhancement greatly improves the model's ability to distinguish between changed and unchanged regions. This boosts the overall performance of CD. The core mechanism of the IDEM is based on the classic self-attention. Although the self-attention mechanism performs well in capturing global context information, it still has limitations when directly applied to extract different information between bi-temporal image features. To overcome this limitation, this study designed the IDEM. As shown in Fig. 5, this module first subjects the features extracted in the previous stage to preliminary processing through a 3×3 convolutional layer. Subsequently, the rearrange function is used to reshape the feature maps into a two-dimensional matrix, from which the respective Q, K, and V tensors are generated. Next, the Q, K, and V tensors of the two temporal branches are rearranged to perform cross-attention calculation, capturing the intertemporal interaction relationships. The calculated attention weights are finally fused with the original features to generate the weighted bi-temporal intermediate features. This process can be expressed by

$$\text{Self_Attention}\left(Q_i, K_j, V_j\right) = \text{softmax}\left(\frac{Q_i K_j^T}{\sqrt{d_i}}\right) V_j \tag{8}$$

$$F^{\text{inter}}_{T1} = F_{T1} + F_{T1}\left(Q_i, K_j, V_j\right) \tag{9}$$

$$F^{\text{inter}}_{T2} = F_{T2} + F_{T2}\left(Q_j, K_i, V_i\right) \tag{10}$$

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where F_{T1} and F_{T2} represent the input bi-temporal feature maps, respectively. F_{T1}^{inter} and F_{T2}^{inter} represent the intermediate enhanced feature maps weighted by the self-attention mechanism, $Q_{i,j} \in R^{H\times W\times C}, K_{i,j} \in R^{H\times W\times C}, V_{i,j} \in R^{H\times W\times C}$,$d_i$represents the dimension of the key vector, and the Softmax(•) function is used to convert the attention scores into a probability distribution.

Subsequently, the enhanced bi-temporal feature maps F_{T1}^{inter} and F_{T2}^{inter} are subtracted to obtain a preliminary difference map F_{diff}, which is fused with the enhanced features. Finally, a 1×1 convolutional layer is used to generate the secondarily enhanced difference feature map. This series of operations aims to preserve the unique advantages and invariant features of each temporal image while enhancing the ability to identify detailed change information. This process can be expressed by

$$F_{\text{diff}} = \left|F_{T1}^{\text{inter}} - F_{T2}^{\text{inter}}\right| \tag{11}$$

$$F_{T1}^{\text{inter}'} = F_{T1}^{\text{inter}} \otimes F_{\text{diff}}{}' \tag{12}$$

$$\mathrm{F}_{T1}^{\text{ide}} = \mathrm{F}_{T1}^{\text{inter}'} \otimes \text{Conv}\left(\text{concat}\left(\mathrm{F}_{\text{diff}}'', F_{T1}^{\text{inter}'}\right)\right) \tag{13}$$

where F_{diff} represents the preliminary difference map obtained by subtraction, $F_{T1}^{\text{inter}'}$ represents the result of fusing the difference feature map, which was secondary enhanced at T1 phase, with the feature map enhanced by the self-attention mechanism, F''_{diff} represents the final difference map obtained after further processing, and F_{T1}^{ide} is the final feature map output by the IDEM module. The calculation process for the T2 branch is similar to that of the T1 branch. To maximize the utilization of different information, this article fuses the final difference feature map with the bi-temporal intermediate features and then subtracts them to obtain a feature map containing significant change region information. Finally, the enhanced difference feature map is fused with the bi-temporal intermediate features. This fusion generates difference-rich bi-temporal feature representations, significantly enhancing the recognition of change regions and improving the model's sensitivity to subtle surface changes in complex scenes, and it also reduces its reliance on domain-specific information. This further assists the model in effectively distinguishing change regions from the background context, providing strong support for the precise final localization of change regions.

D. Loss Function

To achieve intertemporal domain adaptive CD for remote sensing images, this article designs a loss function composed of two parts.

Domain Adversarial Loss: Used to achieve feature distribution alignment, defined as

$$L_{\text{adv}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(D\left(F_{T1}^{i}\right)\right) + (1-y_i)\log\left(1 - D\left(F_{T2}^{i}\right)\right)\right]. \tag{14}$$

D represents the domain classifier, y_i is the domain label of the ith sample, the T1 branch is 0, and the T2 branch is 1. F_{T1}^{i} represents the feature of the ith sample from the T1 branch, F_{T2}^{i} represents the feature of the ith sample from the T2 branch.

CD Loss: Uses binary cross-entropy loss to supervise the final CD result

$$L_{\text{cd}} = -\frac{1}{N}\sum_{i=1}^{N}\left[M_i \log\left(P_i\right) + (1-M_i)\log\left(1-P_i\right)\right]. \tag{15}$$

P_i represents the predicted change probability and M_i is the true change label.

The total loss function is defined as

$$L_{\text{total}} = L_{\text{cd}} + \lambda_{\text{adv}} L_{\text{adv}} + \lambda_{\text{BDL}} L_{\text{BDL}} \tag{16}$$

where λ_{adv} and λ_{BDL} are weighting coefficients that balance the different loss terms.

This multitask loss function considers both domain adaptation and CD needs. The domain adversarial loss aligns the global feature distribution. BDL provides pixel-level, fine-grained supervision. The CD loss ensures accurate final detection results. These three parts work together to improve model performance.

IV. Experiment

A. Datasets

To evaluate the performance of the proposed ADANet-CD network model, this study conducted experiments on three representative remote sensing image CD datasets: CDD, LEVIR-CD, and DSIFN-CD.

1) *CDD [50]:* This dataset is from Google Earth. Its spatial resolution is 0.3–1.0 m. It shows significant seasonal and spectral variations. There are 16 000 image pairs, each 256×256 in size. These are split into training, validation, and testing sets at a 10 000/3000/3000 ratio. It includes various change types like buildings, roads, and vehicles. Its seasonal variations make it good for testing the model's domain adaptation ability.
2) *LEVIR-CD [27]:* This dataset contains bi-temporal remote sensing image pairs from Texas, taken between 2013 and 2020. It has a 0.5-m spatial resolution. It includes 10 000 image pairs, each 1024×1024 in size. These pairs cover many land-cover change types. The training, validation, and testing sets have 7120, 1024, and 1869 image pairs, respectively. Due to different acquisition times, this dataset shows a clear domain shift, making it suitable for testing the model's robustness to seasonal changes.
3) *DSIFN-CD [51]:* This public binary CD dataset was used for experimental validation. It has six pairs of high-resolution (2 m) satellite remote sensing images. These images are from six major cities in China. The DSIFN-CD dataset covers a wide range of land-cover change scenarios. These include changes in roads, buildings, farmland, and water bodies. This provides rich test cases for evaluating CD algorithms. The dataset's diversity helps measure algorithm robustness and generalization across different land-cover types and change scenarios.

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B. Evaluation Metrics

This article uses five common metrics to evaluate the proposed model's performance and accuracy: precision, recall, F1-score, and intersection over union (IoU). CD is a binary classification problem. Each pixel is either classified as changed (P) or unchanged (N). Comparing predictions to ground truth (GT) gives four outcomes: true positive (TP), false positive (FP), false negative (FN), and true negative (TN).

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive. The formula is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

Recall measures the proportion of actual positive samples that are correctly predicted, and its calculation formula is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

F1-score is the harmonic mean of precision and recall. Since there may be a tradeoff between precision and recall, making it difficult to judge the merits of a model, the F1-score is used to comprehensively evaluate the model's performance, and its calculation formula is as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

IoU: Used to measure the degree of overlap between the predicted change area and the actual change area. The higher the IoU value, the greater the overlap between the prediction result and the GT, and the more accurate the model prediction; conversely, the lower the IoU value, the worse the model performance. Its calculation formula is as follows:

$$\text{IoU} = \frac{A \cap B}{A \cup B}.$$

C. Experimental Setup

This study used the PyTorch framework for model development. Experiments on the ADANet-CD network and other methods were run on an NVIDIA GeForce RTX 4070 Ti Super GPU. The network was trained with the Adam optimizer. The initial learning rate was 1e-4, the batch size was 8, and training ran for 150 epochs. For loss function weights, the domain adversarial loss weight was 0.1, and the BDL loss weight was 0.05. A progressive training strategy was used: the feature extractor was warmed up for 10 epochs. Then, domain adaptation training was gradually introduced. A poly learning rate policy dynamically adjusted the learning rate.

D. Hyperparameter Analysis

In the total loss function of ADANet-CD (16), hyperparameters λ_{adv} and λ_{BDL}, respectively, weight the adversarial loss L_{adv} and the BDL L_{BDL}. These hyperparameters are crucial for the model's domain adaptation ability and its final CD performance. To validate the rationality of our chosen hyperparameters and to improve the reliability of our model, we conducted a systematic hyperparameter analysis. We used a controlled variable method. We fixed other experimental settings and examined the impact of each hyperparameter's value on model performance. Specifically, we first fixed λ_{BDL} at 0.05. We then iterated through λ_{adv} values of [0.01, 0.05, 0.1, 0.15,0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5]. Based on this, we fixed λ_{adv} at its optimal value. We then performed a similar iteration for λ_{BDL} with values of [0.01, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3]. We evaluated key metrics such as F1-score and IoU on the validation set to determine the optimal combination. As shown in Fig. 6(a), when λ_{BDL} was fixed at 0.05, the model's F1-score changed with λ_{adv}. The F1-score increased as λ_{adv} grew from 0.01 to 0.1, reaching its peak at $\lambda_{\text{adv}} = 0.1$. The performance began to decline as λ_{adv} increased further. This indicates that a small λ_{adv} is insufficient for effective adversarial domain alignment. However, an excessively large λ_{adv} may cause the model to overemphasize domain invariance, which can harm its ability to discriminate changes or lead to unstable training. After determining the optimal λ_{adv} to be 0.1, we further examined the impact of λ_{BDL}. As shown in Fig. 6(b), with λ_{adv} fixed at 0.1, the F1-score gradually improved as λ_{BDL} increased from 0.01 to 0.05, reaching its best performance at $\lambda_{\text{BDL}} = 0.05$. Performance slightly declined when λ_{BDL} exceeded 0.05. This suggests that a moderate λ_{BDL} effectively guides the feature distance distribution for better CD. However, an overly high λ_{BDL} can impose excessive constraints on the feature learning process, which may limit the model's generalization ability. In summary, through systematic hyperparameter analysis, we identified the optimal hyperparameter combination for ADANet-CD on the adopted datasets as $\lambda_{\text{adv}} = 0.1$ and $\lambda_{\text{BDL}} = 0.05$. This combination effectively optimizes the synergy between learning domain-invariant features and accurately discriminating bi-temporal features. The experimental results not only validate the rationality and robustness of the chosen hyperparameters but also provide a reliable basis for the application of our method in different scenarios.

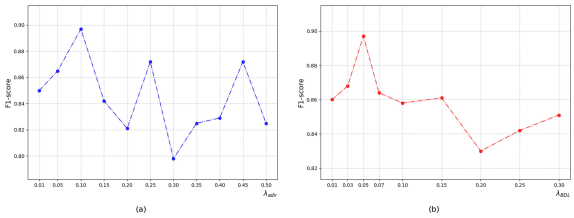


Fig. 6. (a) Impact of λ_{adv} on ADANet-CDś performance, where λ_{BDL} is fixed at 0.05. (b) Impact of λ_{BDL} on ADANet-CDś performance, where λ_{adv} is fixed at 0.1.

E. Comparative Experiments

To verify the effectiveness of ADANet-CD in CD and evaluate its performance, this article selects the remote sensing image CD datasets CDD, DSIFN-CD, and LEVIR-CD, and compares them with methods such as FC-EF, FC-Siam-Diff, FC-Siam-Conc, STANet, BIT, Changeformer, Bifa, and DA-IDANet. The quantitative results, as shown in Table I, demonstrate that on the CDD dataset, precision, F1, and IoU are 2.33%, 2.51%, and 5.29% higher than the current best model, respectively. On the DSIFN-CD dataset, some metrics

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TABLE I
EVALUATION METRICS OBTAINED ON THE CDD, LEVIR-CD, AND DSIFN-CD DATASETS

Methods	CDD				DSIFN-CD				LEVIR-CD			
	Precision	Recall	F1	IoU	Precision	Recall	F1	IoU	Precision	Recall	F1	IoU
FC-EF	83.05	55.56	66.58	49.91	72.61	52.73	61.09	43.98	87.90	62.93	73.35	57.91
FC-Diff	92.27	54.35	68.41	51.98	59.67	65.71	62.54	45.50	89.53	83.31	86.31	75.92
FC-Conc	91.68	64.57	75.77	61.00	66.45	54.21	59.71	42.56	91.99	76.77	83.69	71.96
STA-Net	92.26	84.49	88.20	78.90	67.71	61.68	64.56	47.66	83.81	91.00	87.26	77.40
BIT	92.63	83.91	88.06	78.66	68.36	70.18	69.26	52.97	89.24	89.37	89.31	80.68
ChangeFormer	90.02	83.93	86.87	81.27	88.48	84.94	86.67	76.48	92.05	88.80	90.40	81.27
BiFA	93.15	86.32	89.61	81.17	73.99	68.87	71.34	55.45	91.52	89.86	90.69	82.96
DA-IDANet	92.88	87.05	89.87	80.56	74.28	76.36	75.30	60.39	91.60	73.73	81.70	68.82
WS-Net++	93.65	**89.91**	91.74	84.74	85.24	**86.37**	85.80	75.13	92.57	**90.08**	91.31	84.01
Ours	**95.21**	89.73	**92.38**	**85.85**	**91.17**	84.21	**87.55**	**77.85**	**93.26**	89.78	**91.48**	**84.30**

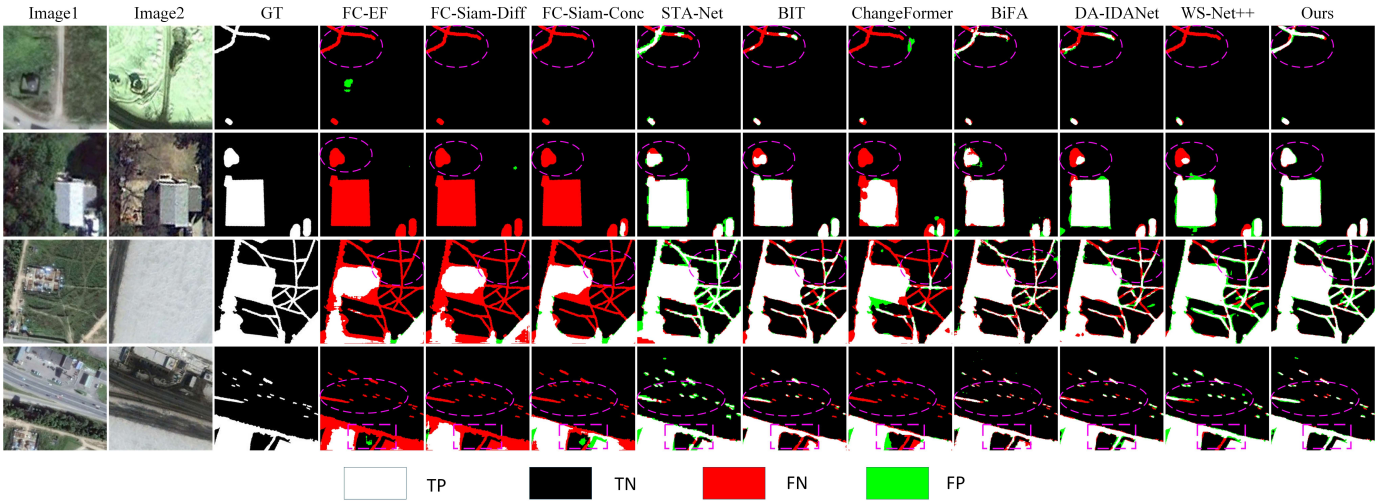


Fig. 7. Visualization results of different methods on the CDD dataset.

are also superior to the current best model, with precision 2.69% higher, F1 0.88% higher, and IoU 1.37% higher. On the LEVIR-CD dataset, all metrics are improved, with precision 1.74% higher, F1 0.79% higher, and IoU 1.34% higher than the current best model.

1) *CDD Experimental Results Visualization:* Fig. 7 shows the four pairs of bi-temporal remote sensing images used. These image pairs were taken in different seasons. They present complex and diverse land-cover changes. Specifically, the first and third rows show road changes in different seasons. The second row shows buildings being added or removed. The fourth row shows small vehicles appearing in the changed area. These bi-temporal image pairs have large seasonal differences, which is a major challenge for CD. Experimental results show that other comparative models are affected by these seasonal changes. This leads to missed detections in changed areas and false detections in unchanged areas. Their accuracy is insufficient, especially for identifying building or road edges. In contrast, the proposed ADANet-CD model effectively aligns features between bi-temporal images. It does this by using a domain adaptation strategy. This significantly reduces interference from seasonal domain shift on CD. Thus, it improves overall CD accuracy.
2) *DSIFN-CD Results Visualization:* Fig. 8 shows the CD results for the DSIFN-CD dataset. This dataset includes many land-cover change types and has high scene com-

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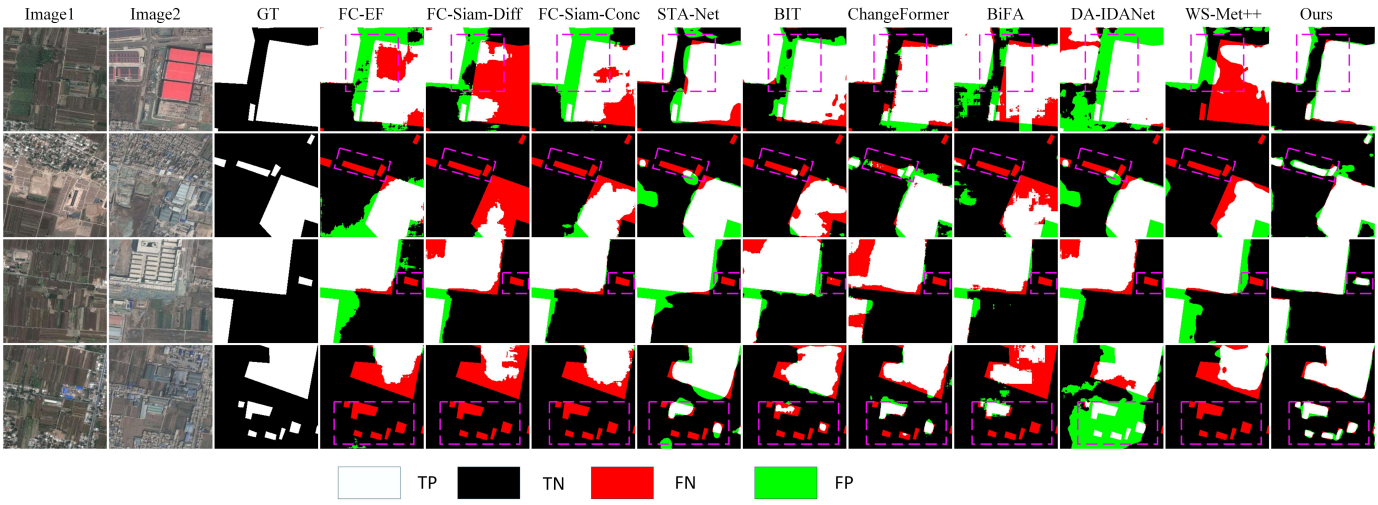


Fig. 8. Visualization results of different methods on the DSIFN-CD dataset.

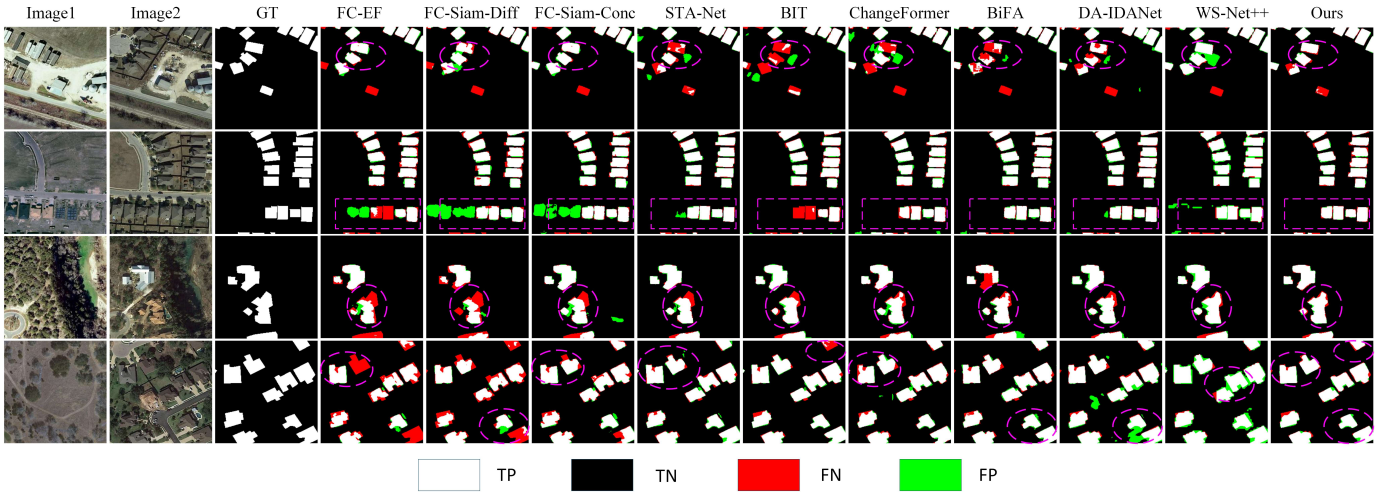


Fig. 9. Visualization results of different methods on the LEVIR-CD dataset.

plexity. As the figure shows, the bi-temporal image pairs in the first to fourth rows have large spectral differences. This increases the difficulty of CD. Experimental results show that the proposed ADANet-CD model helps mitigate such domain shift. It also enhances the model's focus on real change information through an enhanced difference information mechanism. This further proves ADANet-CD's effectiveness in remote sensing image CD tasks with significant domain shift.

3) *LEVIR-CD Results Visualization:* Fig. 9 shows the CD results for the LEVIR-CD dataset. The four pairs of bi-temporal images in this dataset have large spectral differences. The first to third rows mainly show scattered small building changes. In the fourth row, some buildings may be hidden by trees, making building edge identification harder. As the figure shows, ADANet-CD still performs well in handling building edges during CD. It effectively captures subtle change information. This means the method can rely less on domain-specific information, making detection results more complete. It also proves its effectiveness in CD tasks with different illumination conditions.

F. Ablation Study

To verify the importance of each module in the model, we ran several ablation experiments. We removed and swapped module positions to see how effective and logical each module is. Table II shows the results from these experiments on the three datasets.

1) Domain Adaptation Strategy: To see how the GRL and domain classifier affect model performance, we removed them. Table II shows that the model's F1-score on CDD, LEVIR-CD, and DSFIN-CD dropped to 89.49%, 88.72%, and 85.12%, respectively. This is a 2%–4% decrease from the original. This proves that the domain adaptation strategy is very important. Fig. 10, a T-SNE visualization, shows that the domain shift between bi-temporal images greatly improves after applying domain adaptation.

2) TMAM: To evaluate the TMAM's impact, we removed it and ran experiments. Table II shows that without the TMAM,

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TABLE II
COMPARISON OF ABLATION STUDY RESULTS

Method	GRL/DC	TMAM	BDL	IDEM	CDD		LEVIR-CD		DSIFN-CD	
					F1(%) (↓ %)	IoU (↓ %)	F1(%) (↓ %)	IoU (↓ %)	F1(%) (↓ %)	IoU (↓ %)
ADANet-CD	×	✓	✓	✓	89.49 (↓ 2.89)	80.98 (↓ 4.87)	88.72 (↓ 2.76)	79.73 (↓ 4.57)	85.12 (↓ 2.43)	74.09 (↓ 3.76)
	✓	×	✓	✓	90.32 (↓ 2.06)	82.35 (↓ 3.50)	89.67 (↓ 1.81)	81.28 (↓ 3.02)	85.66 (↓ 1.89)	74.92 (↓ 2.93)
	✓	✓	×	✓	91.10 (↓ 1.28)	83.65 (↓ 2.20)	90.44 (↓ 1.04)	82.55 (↓ 1.75)	86.63 (↓ 0.92)	76.41 (↓ 1.44)
	✓	✓	✓	×	90.40 (↓ 1.98)	82.14 (↓ 3.71)	89.96 (↓ 1.52)	81.74 (↓ 2.56)	85.67 (↓ 1.88)	74.93 (↓ 2.92)

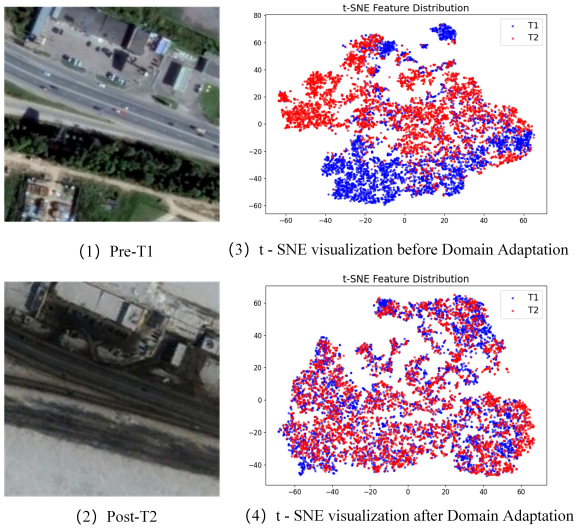


Fig. 10. t-SNE visualization of feature distribution differences before and after domain adaptation of bi-temporal remote sensing images.

TABLE III
COMPUTATIONAL EFFICIENCY IS EVALUATED ON THE CDD DATASET IN TERMS OF PARAMETERS (PARAMS.), FLOPS, AND INFERENCE TIME PER IMAGE PAIR (TIME), USING AN INPUT RESOLUTION OF 256 × 256 × 3 FOR CONSISTENT AND FAIR COMPARISON

Methods	Params(M)	FLOPs(G)	Time(MS)
FC-EF	1.35	3.21	5.37
FC-Siam-Diff	1.35	4.73	10.54
FC-Siam-Conc	1.55	5.33	11.34
BIT	3.49	10.63	29.01
ChangeFormer	51.11	41.88	52.87
BiFA	5.58	53.00	43.08
DA-IDANet	18.47	46.35	22.82
WS-Net++	62.23	31.10	32.99
ADANet-CD	17.52	15.43	25.19

the F1-score on the three datasets were 90.32%, 89.67%, and 85.66%. Fig. 11 compares metrics with and without the TMAM on CDD, DSIFN-CD, and LEVIR-CD. Without the TMAM, F1-score on CDD, LEVIR-CD, and DSIFN dropped by 2.06%, 1.81%, and 1.89%, respectively. This is because a lack of feature interaction between bi-temporal images lowers model performance. This confirms the module's positive effect.

3) BDL: Similarly, we removed the BDL distance guidance constraint. Table II shows that the model's F1-score on CDD, LEVIR-CD, and DSIFN were 91.10%, 90.44%, and 86.63%. This shows that the distance guidance constraint helps the domain adaptation strategy better align features between bi-temporal images. It does this by applying different constraints to changed and unchanged regions. This helps reduce the domain shift problem between bi-temporal images.

4) IDEM: To precisely evaluate the IDEM module's impact on this CD model, we removed it. As Table II shows, without the IDEM, the F1-score on CDD, LEVIR-CD, and DSIFN datasets were 90.40%, 89.96%, and 85.67%. Since the IDEM provides bi-temporal difference features, removing it causes a severe performance drop. This is because the model then lacks enhanced difference map features and contextual information interaction. Fig. 12 shows the visualization results. The experimental results show that the IDEM proposed here effectively enhances change feature representation and improves the model's CD performance.

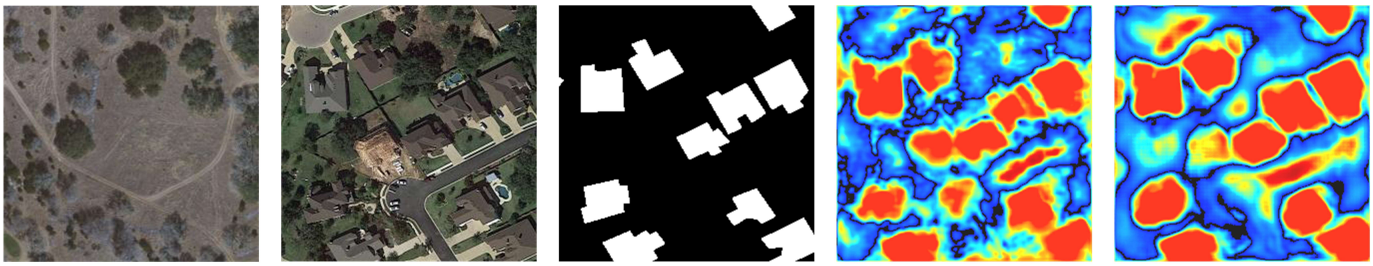
G. Computational Complexity Analysis

The proposed method, referred to as Ours in Table III, demonstrates an excellent balance between model complexity and efficiency. A comparative analysis with existing methods, as detailed in Table III, reveals that our model's parameter count is 17.52M. While this is higher than lightweight models such as FC-EF (1.35M), FC-Siam-Conc (1.55M), BIT (3.49M), and BiFA (5.58M), this increase is justified by the significant performance improvements. Furthermore, our model's parameter count is substantially lower than that of ChangeFormer (51.11M), WS-Net+ + + (62.23M), and

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Fig. 11. (a) Bar chart comparing evaluation metrics with and without the TMAM module on the CDD dataset. (b) Bar chart comparing evaluation metrics with and without the TMAM module on the DSIFN-CD dataset. (c) Bar chart comparing evaluation metrics with and without the TMAM module on the LEVIR-CD dataset.



(1) Pre-T1 (2) Post-T2 (3) Ground Truth (4) Feature visualization before adding the IDEM module (5) Feature visualization after adding the IDEM module

Fig. 12. Heatmap comparison before and after adding the IDEM.

DA-IDANet (18.47M). Regarding computational cost, our model's FLOPs value is 15.43G, which is notably lower than BiFA (53.00G) and DA-IDANet (46.35G), indicating reduced computational resource requirements and lower hardware demands. Additionally, the inference time for a pair of bi-temporal images is 25.19 ms, demonstrating a competitive processing speed that is faster than all evaluated methods except for FC-EF, FC-Siam-Diff, FC-Siam-Conc, and DA-IDANet. These comprehensive metrics collectively highlight that our proposed model achieves an excellent tradeoff between model accuracy and operational efficiency.

Despite a slight increase in parameters (Params) and FLOPs compared to some SOTA models, our model achieves the best accuracy. We conducted a series of rigorous ablation studies to confirm that this performance boost is not from simply stacking parameters. It stems from the genuine effectiveness and sophisticated design of each core module. Specifically, our ablation experiments were designed to systematically evaluate each module's contribution to overall performance. The results in Table II clearly show that removing any key module (GRL/dc, TMAM, BDL, and IDEM) leads to a significant and quantifiable performance drop. For instance, removing GRL and the domain classifier resulted in a 2.89% decrease in the F1-score, while removing the IDEM led to a 2.56% drop in IoU. These data strongly prove the indispensability and unique contribution of each module to the model's final excellent performance, rather than a cumulative effect of redundant parameters. This shows that our model achieves a notable balance between efficiency and performance. Its excellent performance is a result of the synergistic effect and optimized design of its components. Our research focuses on maximizing performance gains through carefully designed modules with minimal additional computational cost, rather than blindly increasing model size.

V. Conclusion

This study presents a new domain adaptation-based method for bi-temporal remote sensing image CD. Its goal is to effectively reduce the domain shift problem caused by feature distribution differences between image pairs and improve CD accuracy. Specifically, in the encoding stage, a domain classifier is placed on the T1 branch. An adversarial training framework is built between the feature extractor and the domain classifier using a GRL. This encourages the feature extractor to learn domain-invariant feature representations. To further improve the domain classifier's discriminative ability for bi-temporal features, this study proposes a TMAM. This module extracts more distinctive temporal feature representations, providing reliable inputs for the GRL and domain classifier. Additionally, this study designs a BDL. This loss explicitly regulates the distance between change and non-change region features in the feature space. This further promotes the alignment of bi-temporal features, more effectively addressing the domain shift problem. In the decoding stage, this study develops an IDEM. This module aims to increase the model's focus on change information and reduce its reliance on domain-specific information. This achieves accurate identification of change regions. To fully confirm

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the method's effectiveness, extensive comparative experiments were run on three public remote sensing image CD datasets: CDD, DSIFN-CD, and LEVIR-CD. The results show that the proposed method significantly improves performance on all evaluation datasets, outperforming existing mainstream methods. In the future, we will focus on several advanced research directions. First, given the diversity of remote sensing data modalities, we will explore extending the ADANet-CD framework. We aim to move from our current "temporal domain" adaptation to the more challenging "cross-sensor domain" adaptation. This includes the CD between optical and SAR images. It requires us to design more robust feature alignment strategies to handle significant differences between sensor modalities. Furthermore, we will investigate applying our domain adaptation mechanism to "cross-region domain" adaptation. This addresses problems across different geographical contexts. The goal is to improve the model's generalization and adaptability when facing data from untrained geographical areas. Finally, to reduce the reliance on large amounts of labeled data, we will actively explore few-shot or unsupervised domain adaptation strategies. This will allow ADANet-CD to maintain efficient domain adaptation even when labeled samples are scarce in the target domain. This will significantly expand its applicability in real-world remote sensing applications.

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